# Internal Relay Instructions

Modular Logic with Internal Storage Bits

## 1 Internal Storage Bits

PLCs allocate a specific area of memory for **Internal Storage Bits**. These bits are essential for creating complex logic without being limited by physical hardware.

**Terminology**

Internal storage bits are also commonly referred to as:

- Internal Outputs or Internal Coils.
- Internal Control Relays.
- Internal Bits or Storage Points.

## 2 The Role of Internal Outputs

Internal outputs are ON/OFF signals generated solely by the programmed logic. They serve as "virtual relays" within the processor's memory.

- **No Physical Connection:** They do not directly control any output field device (like a motor or lamp). No wiring is connected to these addresses.
- **Internal Logic Purpose:** They are used to store intermediate logic states that can be referenced elsewhere in the program.
- **Resource Conservation:** They perform relay functions without occupying expensive physical output points on the PLC modules.

## 3 Addressing: SLC 500 Bit File (B3)

For Allen-Bradley **SLC 500** controllers, the dedicated memory file for internal storage is the **Bit File (B3)**.

**Addressing Format**

The address for an internal bit starts with the file identifier, followed by the word and bit number:

```
B3 :  [Word] / [Bit]
```

**Example:** `B3:1/3` indicates File 3 (Bit File), Word 1, Bit 3.

## 4 Application: Internal Control Relays

One of the most practical uses for internal relays is to overcome **Matrix Limitations** (hardware constraints on the number of instructions per rung).

### 4.1 Solving Series Contact Limits

If a rung is limited to **7 series contacts** but the required logic demands **12 series contacts**, an internal relay provides the solution.

**Rung Splitting Strategy**

1. **Rung 1:** Program the first 6 contacts to drive an intermediate **Internal Bit** (e.g., `B3:0/0`).
2. **Rung 2:** Use the "status" of the internal bit (`XIC B3:0/0`) in series with the remaining 6 contacts to drive the final physical output.

**Summary**

Internal relays act as the "scaffolding" of a PLC program, allowing for infinite logical complexity by bridging rungs and storing temporary states safely in memory.

*Optimize your logic, conserve your hardware.*